



Pablo Villacorta Aylagas

Telecommunications Engineer
PhD Candidate

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May 29, 2000

Valladolid, Spain



About me

I am a Telecommunications Engineer interested in all its areas, from signal processing to programming and telematics, including disciplines such as electronics and instrumentation. I am also passionate about Magnetic Resonance Imaging (MRI), a field that brings together the best of each discipline and offers room for innovation to improve people's health.

Currently, I am working as a researcher in the Image Processing Lab (LPI) at Universidad de Valladolid (Spain), where I contribute to the **KomaMRI.jl** project while developing a **web-based tool for MRI pulse sequence editing and simulation**.

Resume

Experience

MRI Researcher – PhD Candidate
Image Processing Lab (LPI) – Universidad de Valladolid

2022 – Present

Working on MRI simulation frameworks, with a focus on modeling complex phenomena like blood flow and non-rigid motion. Also involved in optimizing simulation pipelines and improving algorithms performance and accuracy.

Education

Universidad de Valladolid, Spain
MSc in Telecommunications Engineering
2022 – 2023

Habilitating MSc focused on real-world projects. We designed an IoT datalogger with LoRa and sensor integration, and deployed a 4G-based system for real-time video processing and AI traffic sign detection.

Universidad de Valladolid, Spain
BSc in Telecommunications Engineering
2018 – 2022

Acquired a solid foundation in signal processing, programming, electronics, systems engineering, and telematics.

Languages

English – C1 Level
Certified with Cambridge Certificate in Advanced English (CAE)

2026

Score: 198

Certified with Cambridge First Certificate in English (FCE)

2020

Score: 182

Publications

Journal Papers

2026

- MRSeqStudio: MRI Sequence Design and Simulation *as a Service* in a Free and Open-Source Web Platform
J Med Syst, 50, 65 (2026). DOI: 10.1007/s10916-026-02394-1
P. Villacorta-Aylagas, M. Rodríguez-Cayetano, C. Castillo-Passi, P. Irarrazaval, F. Simmross-Wattenberg, C. Alberola-López

2025

- Versatile and Highly Efficient MRI Simulation of Arbitrary Motion in KomaMRI
Magn Reson Med, 95, no. 3 (2026): 1791–1803. DOI: 10.1002/mrm.70145
P. Villacorta-Aylagas, C. Castillo-Passi, R. Kierulf, R.M. Menchón-Lara, J.R. Rodríguez-Galván, J. Sierra-Pallaers, P. Irarrazaval, C. Alberola-López
- Sampling of non-Gaussian Ensemble Average Propagators for the simulation of diffusion magnetic resonance images
Magn Reson Med, 95, no. 2 (2026): 1255–1265. DOI: 10.1002/mrm.70080
J.R. Rodríguez-Galván, P. Villacorta-Aylagas, S. Merino-Caviedes, F. Simmross-Wattenberg, C. Castillo-Passi, P. Irarrazaval, A. Tristán-Vega, C. Alberola-López

Conference Contributions

2026

- Enhancing KomaMRI with a Pulseq I/O Interface for Reproducible Sequence Workflows
2026 ISMRM Iberian Chapter Meeting, Coimbra, Portugal
P. Villacorta-Aylagas, C. Castillo-Passi, P. Irarrazaval, F. Simmross-Wattenberg, M. Rodríguez-Cayetano, C. Alberola-López
[Abstract] [Poster]
- MRSeqStudio: A Free and Open-Source Web Application for MR Sequence Design and Simulation as a Service
2026 ISMRM & ISMRT Annual Meeting & Exhibition, Cape Town, South Africa
P. Villacorta-Aylagas, C. Castillo-Passi, P. Irarrazaval, F. Simmross-Wattenberg, M. Rodríguez-Cayetano, C. Alberola-López

2025

- A Free and Open-Source Web Application for Pulse Sequence Development and Simulation
🏆 1st Best Poster Presentation 🏆
2025 ISMRM Iberian Chapter Meeting, Barcelona, Spain
P. Villacorta-Aylagas, C. Castillo-Passi, P. Irarrazaval, F. Simmross-Wattenberg, M. Rodríguez-Cayetano, C. Alberola-López
[Abstract] [Poster]

2024

- Flexible Motion Models for GPU-accelerated MRI Simulations with Complex Motion and Flow using KomaMRI
ESMRMB 2024, Barcelona, Spain
P. Villacorta-Aylagas, C. Castillo-Passi, R.M. Menchón-Lara, J. Anatol-Hernández, P. Irarrazaval, J. Sierra-Pallares, C. Alberola-López
[Abstract] [Poster]
- Proof of Concept for the Simulation of Diffusion Models in the Myocardium through KomaMRI
CASEIB 2024, Sevilla, Spain
J. R. Rodríguez-Galván, P. Villacorta-Aylagas, S. Merino-Caviedes, C. Castillo, J. Sánchez-González, D. Filgueiras-Rama, A. Tristán-Vega, P. Irarrazaval, C. Alberola-López

- **New Methods for Defining Dynamic Phantoms in KomaMRI**
2024 ISMRM Iberian Chapter Meeting, Porto, Portugal
P. Villacorta-Aylagas, C. Castillo-Passi, R.M. Menchón-Lara, J. Anatol-Hernández, P. Irarrazaval, J. Sierra-Pallares, C. Alberola-López
- **Synthetic spin trajectory generation for advanced Flow phantom in MRI simulation**
2024 ISMRM Iberian Chapter Meeting, Porto, Portugal
J. Anatol-Hernández, J. Sierra-Pallares, **P. Villacorta-Aylagas**, C. Castillo-Passi, R.M. Menchón-Lara, P. Irarrazaval, C. Alberola-López
- **A KomaMRI Phantom Extension for Integrated Dynamic Imaging**
2024 ISMRM & ISMRT Annual Meeting & Exhibition, Singapore
P. Villacorta-Aylagas, C. Castillo-Passi, R. M. Menchón-Lara, P. Irarrazaval, J. B. Sierra-Pallares, C. Alberola-López

2023

- **A Web Version of KomaMRI. Sequence Editor and Remote Execution**
2023 ISMRM Iberian Chapter Meeting, Valladolid, Spain
P. Villacorta-Aylagas, I. Fernández-Arias, C. Castillo-Passi, P. Irarrazaval, F. Simmross-Wattenberg, M. Rodríguez Cayetano, C. Alberola-López
- **Parallel Device-Agnostic Simulation of Magnetic Resonance Imaging Sequences**
2023 ISMRM Iberian Chapter Meeting, Valladolid, Spain
I. Fernández-Arias, **P. Villacorta-Aylagas**, C. Castillo-Passi, P. Irarrazaval, F. Simmross-Wattenberg, M. Rodríguez-Cayetano, C. Alberola-López

2022

- **A Cross-Platform Editor and Simulator of Magnetic Resonance Imaging Sequences. Design and Implementation**
CASEIB 2022, Valladolid, Spain
P. Villacorta-Aylagas, I. Fernández-Arias, C. Castillo-Passi, P. Irarrazaval, F. Simmross-Wattenberg, M. Rodríguez Cayetano, C. Alberola-López
- **Simulación paralela multidispositivo de secuencias de imagen por resonancia magnética**
CASEIB 2022, Valladolid, Spain
I. Fernández-Arias, **P. Villacorta-Aylagas**, C. Castillo-Passi, P. Irarrazaval, F. Simmross-Wattenberg, M. Rodríguez-Cayetano, C. Alberola-López